

MATH GU4053 INTRO TO ALGEBRAIC TOPOLOGY
HOMEWORK 2

DUE: THURSDAY, FEBRUARY 11, 2026, BEFORE 23:59PM

SUBMIT THE COMPLETED ASSIGNMENT TO <https://www.gradescope.com>

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- This assignment has **5 questions** on **two pages**.
 - This assignment counts towards 2% of your final grade (up to a maximum of 10% for all homework).
 - Your solutions must be *handwritten*. For instance on a tablet/iPad, or with pen and paper scanned to a pdf.
 - While collaborative work is allowed and generally beneficial, you must write up the solutions *on your own* and *in your own words*.
 - Rigor, clarity, and proper use of mathematical notation count! The work you submit for evaluation will require showing all steps and a clear explanation of what you have done. The explanations need not be lengthy to be clear.
 - Solutions based on an advanced result which has not been covered in this class, or one of its prerequisites, will not receive full credit. (Especially if the use of an advanced result significantly changes the difficulty of the problem.)
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1. Let $\mathbb{R}P^n = S^n / \sim$, where $\sim$ is the equivalence relation generated by $x \sim -x$ for $x = (x_0, \dots, x_n) \in S^n \subset \mathbb{R}^{n+1}$. Write $[(x_0, \dots, x_n)] \in \mathbb{R}P^n$ for the equivalence class of x , and let

$$U = \mathbb{R}P^n \setminus \{[(0, \dots, 0, 1)]\}.$$

Prove that the map

$$\begin{aligned} \mathbb{R}P^{n-1} &\rightarrow U \\ [(x_0, \dots, x_{n-1})] &\mapsto [(x_0, \dots, x_{n-1}, 0)] \end{aligned}$$

is a homotopy equivalence.

2. (a) Let $p : X \rightarrow Y$ be a covering. Prove that for any open subset $U \subset Y$, the restriction

$$p|_{p^{-1}(U)} : p^{-1}(U) \rightarrow U$$

is a covering.

- (b) True or false: for any open subset $V \subset X$, the restriction

$$p|_V : V \rightarrow p(V)$$

is a covering map.

Provide either a proof, or a counter-example with justification.

3. Let $G = \mathbb{Q}$ denote the set of rational numbers, considered as a group with respect to addition. Let $X = \mathbb{R}$ have the standard topology, and let

$$\alpha : G \times X \rightarrow X$$

be defined by $\alpha(r, t) = r + t$. It may be used without proof that this defines a continuous group action.

Is it a covering space action? *Justify your answer.*

4. (a) Prove that if $k_0, k_1 : I \rightarrow I$ are two continuous maps with $k_0(0) = k_1(0)$ and $k_0(1) = k_1(1)$, then $k_0 \simeq k_1 \text{ rel } \partial I$.
 (b) Read [GW, Lemma 1.1.9], and use it to deduce that if $k : I \rightarrow I$ is continuous and if $k(0) = 0$ and $k(1) = 1$, then $\lambda \simeq \lambda \circ k \text{ rel } \partial I$ for any continuous $\lambda : I \rightarrow X$.
 (c) Deduce that if $k : I \rightarrow I$ is continuous and if $k(0) = k(1) = 0$, then $\lambda \circ k \simeq c_x \text{ rel } \partial I$, for any $\lambda : I \rightarrow X$ with $\lambda(0) = x$.

5. Let $Y = \mathbb{R}/\mathbb{Z}$ be the quotient of $\mathbb{R}$ by the equivalence relation $x \sim x'$ when $x - x' \in \mathbb{Z}$, considered as a based space with basepoint $y = [0]$. Let $g = [\omega] \in \pi_1(Y, y)$ be defined the class represented by the loop $\omega : I \rightarrow Y$ defined as $\omega(t) = [t]$.

Prove that for any based space (Z, z) and any element $g' \in \pi_1(Z, z)$, there exists a based map $f : (Y, y) \rightarrow (Z, z)$ with

$$f_*(g) = g'.$$